GLDMamba: Global–Local Differential Mamba Network for Hyperspectral Image Change Detection

Zhenyang Xie, *Graduate Student Member, IEEE*, Li Lv, Hongmin Gao, *Senior Member, IEEE*, Shufang Xu, *Member, IEEE*, Jeng-Shyang Pan, *Senior Member, IEEE*, Wenping Chen, and Longzhe Han

***Abstract*—Convolutional neural networks (CNNs) and transformer-based methods have achieved significant progress in hyperspectral image (HSI) change detection (CD). However, most existing approaches adopt patch-based frameworks, which limit the effective utilization of full bitemporal HSI information. Moreover, CNNs struggle to capture long-range dependencies, while transformers incur high computational complexity. To address these limitations, this article proposes a global–local differential Mamba (GLDMamba) network for HSI-CD, leveraging the linear complexity and strong global modeling capability of the state space model (SSM)-based Mamba architecture. The proposed method directly processes image-level bitemporal HSI data and consists of three components: a global differential sensing module (GDSM), a local differential sensing module (LDSM), and a feature fusion module (FFM). A bitemporal difference image and a bitemporal concatenation image are constructed as complementary inputs. The GDSM employs a temporal-adaptive 2-D-selective scan (TA-SS2D) mechanism to aggregate global contextual features along four directions, while the LDSM captures localized change information using directional differential convolution kernels with adaptive fusion. Finally, global and local features are integrated through the FFM via cross-differential fusion. Experiments on three public HSI-CD datasets demonstrate that GLDMamba outperforms several state-of-the-art deep learning (DL) methods with fewer parameters and lower runtime. The implementation code is publicly available at https://github.com/RogsXie/GLDMamba**

***Index Terms*—Change detection (CD), global–local difference fusion, hyperspectral image (HSI), image-level processing, Mamba.**

Received 12 March 2026; revised 7 April 2026, 5 May 2026, and 2 June 2026; accepted 6 June 2026. Date of publication 9 June 2026; date of current version 18 June 2026. This work was supported by the National Natural Science Foundation of China under Grant 62066030. *(Corresponding authors: Li Lv; Hongmin Gao.)*

Zhenyang Xie, Li Lv, Wenping Chen, and Longzhe Han are with the School of Information Engineering, Jiangxi University of Water Resources and Electric Power, Nanchang 330099, China, and also with Jiangxi Province Engineering Research Center for Intelligent Processing and Early Warning Technology of Water Conservancy Big Data, Nanchang 330099, China (e-mail: Xiewak@163.com; lvli@juwp.edu.cn; 2025995050@juwp.edu.cn; lzhan@nit.edu.cn).

Hongmin Gao and Shufang Xu are with the College of Computer Science and Software Engineering, Hohai University, Nanjing 211100, China (e-mail: gaohongmin@hhu.edu.cn; xushufang@hhu.edu.cn).

Jeng-Shyang Pan is with the School of Artificial Intelligence, Nanjing University of Information Science and Technology, Nanjing, Jiangsu 210044, China (e-mail: jspan@ieee.org).

Digital Object Identifier 10.1109/TGRS.2026.3701799

I. Introduction

With the continuous advancement of spectral imaging technologies, hyperspectral images (HSIs) have been increasingly employed in Earth observation applications [1]. Unlike conventional remote sensing imagery, HSIs provide dense spectral sampling over a wide range of wavelengths [2], enabling the acquisition of continuous and highly discriminative spectral signatures. Such rich spectral resolution allows HSIs to effectively characterize complex land-cover dynamics and subtle material variations.

Change detection (CD) constitutes a fundamental research problem in remote sensing, focusing on the identification of surface variations through the analysis of multitemporal observations acquired at different time instances. Benefiting from the superior spectral resolution of HSIs, HSI-CD has demonstrated enhanced capability in capturing temporal variations and spatial heterogeneity. As a result, HSI-CD has been widely applied in various scenarios, including forest conservation [3], agricultural monitoring [4], land-cover analysis [5], and disaster assessment [6].

Traditional HSI-CD methods primarily rely on handcrafted features and statistical measures to identify differences between bitemporal HSIs and can be broadly categorized into algebra-based [7], transformation-based [8], [9], and classification-based methods [10]. These approaches typically depend on predefined spectral or spatial descriptors, which limits their representation capacity and makes them sensitive to noise and spectral variability, thereby resulting in limited robustness. With the advancement of deep learning (DL), convolutional neural network (CNN)-based models, such as Siamese networks [11], U-Net [12], and their variants [13], [14], have significantly improved detection accuracy by learning hierarchical spatial–spectral features. However, due to the inherent locality of convolution operations, their ability to model long-range dependencies and global contextual information remains limited. To address this issue, transformer-based models [15], [16] have been introduced to capture global contextual relationships via self-attention mechanisms, effectively reducing false alarms and missed detections. Consequently, hybrid architectures [17], [18], [19] have been proposed to integrate the strengths of CNNs and transformers, achieving a better balance between local detail representation and global dependency modeling.

1558-0644 © 2026 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:42:48 UTC from IEEE Xplore. Restrictions apply.

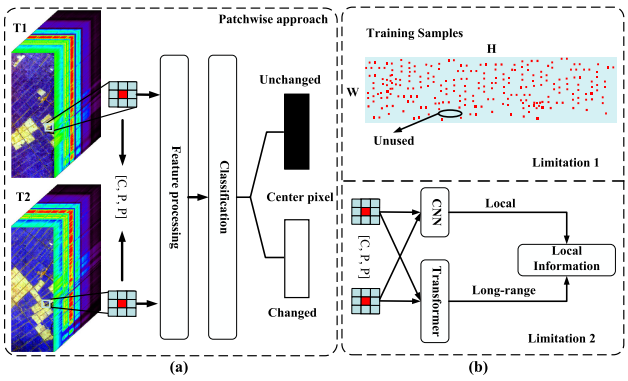


Fig. 1. Patch-based processing and its limitations. (a) Patch-based framework. (b) Limitations.

Despite significant progress in HSI-CD, several critical challenges remain. As illustrated in Fig. 1, most existing methods still rely on a patch-based training paradigm, where local patches are cropped from bitemporal HSIs to learn spatial–spectral–temporal feature representations, and the central pixel is used for change classification. Although this strategy reduces computational cost to some extent, it introduces notable limitations. On the one hand, from the perspective of data utilization, only partially sampled local regions participate in training, leaving a large proportion of hyperspectral pixels underutilized. On the other hand, from a modeling perspective, whether based on convolutional neural networks for local modeling, transformers for global modeling, or their hybrid variants, the effective receptive field is inherently constrained by the patch size, thereby limiting the ability to capture long-range dependencies and holistic change patterns in complex scenes. Essentially, HSI-CD requires the joint modeling of fine-grained local variations and large-scale global change patterns, which cannot be fully achieved within conventional patch-based frameworks. Although recent studies attempt to alleviate this issue by introducing global modeling mechanisms or designing hybrid architectures, these approaches often incur high computational complexity. Furthermore, they generally lack effective decoupling and coordination between global dependency modeling and local feature extraction, and fail to adequately model the interactions between them.

Benefiting from the ability to capture long-range dependencies with low computational cost, state-space model (SSM)-based architectures, particularly Mamba [20], [21], have recently attracted increasing attention. Its capability of modeling long-range dependencies with linear computational complexity makes it particularly suitable for large-scale HSI data processing [22], [23]. Accordingly, several Mamba-based image-level HSI analysis methods have been developed for tasks such as classification, unmixing, and denoising [24], [25], [26]. In the field of HSI-CD, Wu et al. [27] introduced a spatial–spectral Mamba interaction fusion (SSMIF) network, while Li et al. [28] proposed the GASSM model by integrating global attention with Mamba. However, these methods still adopt a patch-based paradigm and fail to fully exploit the advantage of Mamba in modeling long-range dependencies for image-level HSI-CD. To fully utilize all available HSI data and effectively integrate global and local contextual dependencies, we propose a global–local differential Mamba (GLDMamba) network specifically designed for image-level HSI-CD. The proposed GLDMamba consists of three main components: a global differential sensing module (GDSM), which captures long-range dependencies and global change structures across the entire image; a local differential sensing module (LDSM), which precisely models subtle local variations and boundary differences; and a feature fusion module (FFM), which effectively integrates global and local features, emphasizes critical change regions, and enhances the overall discriminative capability.

The main contributions of this work are summarized as follows.

1) A dual-branch HSI-CD framework, termed GLDMamba, is proposed to exploit bitemporal HSIs via differential and concatenation views. By leveraging the sequence modeling capability of Mamba, it enables efficient change representation from full-scene inputs with reduced redundancy, while maintaining strong performance under limited training samples.
2) A global–local bidirectional enhancement strategy is introduced. The GDSM extends the multi-directional 2-D selective scanning mechanism to better model long-range dependencies and global semantics, while the LDSM utilizes directional differential convolutions with adaptive fusion to capture fine-grained local structures, enabling accurate HSI-CD with a lightweight design.
3) A differential-guided FFM is developed to adaptively integrate global and local representations according to change intensity, highlighting discriminative regions. The proposed framework achieves superior performance on three public HSI-CD datasets.

The organization of this article is as follows. Section II reviews existing methods for HSI-CD, including traditional approaches and DL-based techniques. Section III presents the detailed architecture of the proposed GLDMamba. Section IV reports extensive experimental results on three benchmark datasets. Finally, Section V concludes this article.

II. Related Work

A. Traditional HSI-CD Methods

Traditional HSI-CD methods primarily rely on handcrafted features and statistical metrics to identify changes between bitemporal images. Representative techniques include change vector analysis (CVA) [29], which detects changes by measuring spectral differences and applying thresholds, as well as principal component analysis (PCA) [30] and multivariate alteration detection (MAD) [31], which project high-dimensional data into more discriminative subspaces. Variants such as sequential spectral CVA (S2-CVA) [32] further improve detection performance through multilevel analysis. In addition, iterative reweighted MAD (IR-MAD) [33] enhances CD by iteratively refining projections based on canonical correlation analysis.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:42:48 UTC from IEEE Xplore. Restrictions apply.

Some methods further incorporate conventional classifiers, such as support vector machines (SVMs) [34] and maximum likelihood classifiers, to perform CD on transformed or difference features. Other approaches, such as the ultrafusion method [35], model input images as stochastic processes and generate segmentation results via statistical fusion strategies. Despite their simplicity and computational efficiency, these traditional methods are limited by their reliance on manually designed features and insufficient representation capacity, which restrict their effectiveness in complex hyperspectral scenarios.

B. DL-Based HSI-CD Methods

DL has significantly advanced HSI-CD by enabling the automatic extraction of discriminative spatial–spectral representations. CNNs have been widely utilized owing to their strong capability in feature learning and spatial modeling [36], [37]. For instance, Wang et al. [38] proposed a 2D-CNN to extract high-level features and improve generalization performance, while Zhan et al. [11] introduced a Siamese CNN with shared parameters to mitigate temporal discrepancies. Huang et al. [39] developed a hybrid 2-D–3-D CNN framework to enhance representation capability. In addition, encoder–decoder architectures have also been explored to learn multilevel spatial–spectral features [40]. Despite these advances, CNN-based methods are inherently constrained by their local receptive fields, limiting their ability to capture global contextual information. Consequently, in complex remote sensing scenarios with sparse changes or abundant pseudo changes, CNN-based approaches still exhibit considerable room for improvement.

To further enhance global modeling capability, transformer-based methods have attracted increasing attention due to their ability to model long-range dependencies. Ding et al. [41] proposed a transformer encoder-based framework, termed CDFormer, for HSI-CD, while Wang et al. [42] introduced SSTFormer to jointly model spatial, spectral, and temporal dependencies. Furthermore, hybrid architectures combining CNNs and transformers have been explored to leverage their complementary strengths. For example, Shafique et al. [17] integrated CNNs with transformers for efficient feature learning, and Zhang et al. [18] developed a dual-branch Siamese spatial–spectral transformer network (DBS3TAN). To better capture both global and local representations, Wang et al. [43] proposed a global–local attention transformer (GLAFormer). In addition, image-level modeling methods, such as GlobalMind [44], further enhance global representation learning from a holistic perspective. Despite these advances, transformer-based methods generally suffer from high computational complexity due to the quadratic cost of self-attention. Additionally, some existing approaches still rely on predefined and fixed feature fusion strategies, which limit their ability to adaptively balance local details and global semantics across diverse scenarios.

SSMs have attracted increasing attention due to their capability to efficiently model long sequences with linear computational complexity. Mamba [20] represents a significant advancement in this domain by introducing a selective scanning mechanism that dynamically adapts model parameters to the input content, thereby overcoming the content-insensitivity limitation of traditional SSMs. To extend Mamba to vision tasks, VMamba [21] generalizes 1-D sequence modeling to 2-D feature maps by performing bidirectional scanning along horizontal and vertical axes, enabling effective global context aggregation. Building upon this paradigm, Wu et al. [27] proposed the SSMIF network, while Li et al. [28] developed the GASSM model by incorporating a global attention mechanism into Mamba. However, these methods are primarily based on patch-level representations, which limit the ability of Mamba to fully exploit long-range dependencies in image-level CD tasks. In addition, they typically model either difference-based or concatenation-based features while overlooking the complementary relationship between these two representations. More importantly, existing Mamba-based methods generally adopt the standard SS2D without adapting the state space formulation to bitemporal inputs. To address these limitations, we propose a GDSM based on a temporal-adaptive 2-D-selective scan (TA-SS2D) and an LDSM based on differential convolution. These modules perform complementary global and local modeling along two parallel branches, namely the difference branch and the concatenation branch. Their interactions are guided by a FFM, enabling the extraction of discriminative change representations from bitemporal images.

III. Methodology

A. Overview of the Proposed GLDMamba

The overall framework of the proposed GLDMamba is illustrated in Fig. 2(a). The network directly processes bitemporal HSIs as inputs and adopts a dual-branch representation strategy based on difference and concatenation operations. Given a pair of bitemporal HSI data $x_{t1} \in R^{C\times H\times W}$ and $x_{t2} \in R^{C\times H\times W}$, an element-wise subtraction is first performed to generate the difference map $D \in R^{C\times H\times W}$, where the absolute difference is used to emphasize change magnitude while removing directional ambiguity, leading to more stable and discriminative representations. In parallel, a concatenation operation is applied to obtain the stacked representation $C \in R^{2C\times H\times W}$, which is designed to capture latent change patterns and contextual dependencies between the two temporal observations. These operations are formulated as

$$\begin{aligned} D &= |x_{t2} - x_{t1}| \\ C &= \text{Concat}\,(x_{t2}, x_{t1})\,. \end{aligned} \tag{1}$$

To mitigate the increased computational burden and training cost caused by standard 2-D CNN architectures in spatial feature extraction, depthwise separable convolutions are employed for initial feature embedding. Specifically, both the difference map D and the concatenated map C are fed into depthwise separable embedding layers to obtain their aligned feature representations F_D and F_C, respectively

$$\begin{aligned} F_D &= f_{\text{DWConv1}(D)} \\ F_C &= f_{\text{DWConv2}(C)}. \end{aligned} \tag{2}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:42:48 UTC from IEEE Xplore. Restrictions apply.

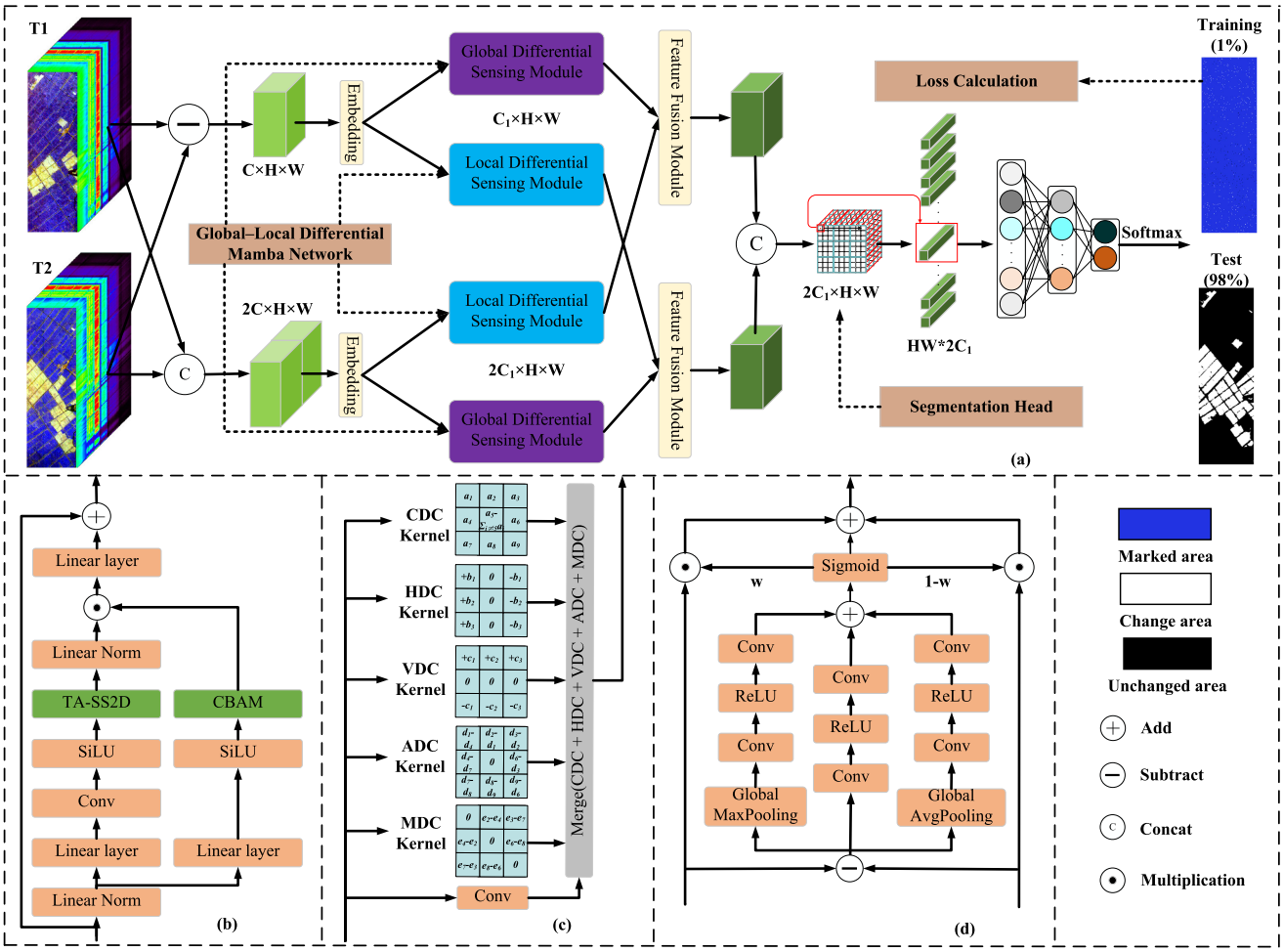


Fig. 2. Overview of the proposed GLDMamba framework for image-level HSI-CD. The final output is a change map, where white pixels represent the changed areas. (a) GLDMamba framework. (b) GDSM. (c) LDSM. (d) FFM.

Subsequently, the features from the two branches are cross-propagated through the FFM, enabling interleaved global–local interaction. The resulting fused features, denoted as F_{out1} and F_{out2}, are then aggregated and fed into a fully connected layer to generate the final prediction map $\widehat{Y} \in R^{2\times H\times W}$. This process can be expressed as

$$\begin{aligned} F_{out} &= \text{Concat}\left(F_{out1}, F_{out2}\right) \\ \widehat{Y} &= \text{Softmax}\left(FC\left(F_{out}\right)\right). \end{aligned} \tag{3}$$

Finally, the CD map is obtained by assigning each pixel to the class corresponding to the highest predicted probability.

B. Global Differential Sensing Module

To model long-range dependencies and enable global differential awareness, a GDSM is introduced, as shown in Fig. 2(b). The GDSM establishes pixel-wise associations through a TA-SS2D mechanism, as illustrated in Fig. 3. By scanning the feature map along four distinct directional paths, the GDSM establishes global dependency relationships between each pixel and the remaining pixels within the scene.

The GDSM consists of three parallel branches: a TA-SS2D branch for modeling long-range dependencies, a multiplicative interaction branch for feature modulation, and a residual branch for information preservation. In the TA-SS2D branch, the feature map is scanned along four traversal paths to generate sequential representations. Specifically, S_{type1} corresponds to scanning from the top-left to the bottom-right in a row-wise manner. During horizontal scanning from left to right, when the last element of a row is reached, the scanning proceeds vertically downward to the next row and resumes from the leftmost element, until the bottom-right pixel is reached. S_{type2} denotes column-wise scanning from top to bottom. When the last element of a column is reached, the scanning moves to the next column on the right and starts again from the top element. S_{type3} and S_{type4} follow the reverse traversal orders of the above two directions.

In the TA-SS2D branch, to overcome the limitation that the standard SS2D employs identical discretization parameters across all spatial positions and scanning directions without differentiating between changed and unchanged regions, a direction-decoupled, drift-aware mechanism is introduced. This mechanism adaptively modulates the global state

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:42:48 UTC from IEEE Xplore. Restrictions apply.

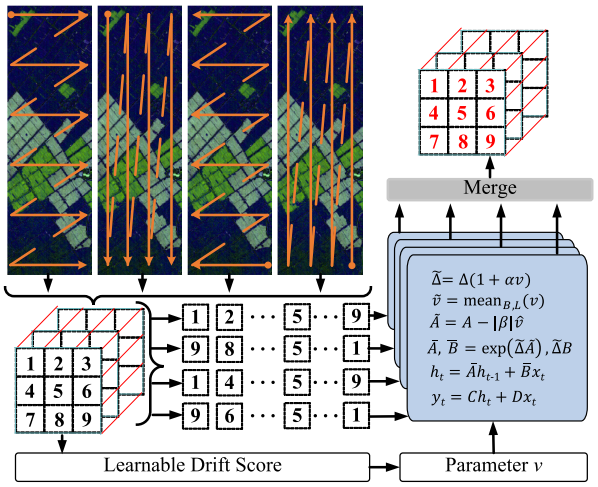


Fig. 3. TA-SS2D operation. The input feature map is scanned along four directions—(left-to-right), (right-to-left), (top-to-bottom), and (bottom-to-top)—to generate directional sequential representations. These sequences are then processed to capture long-range dependencies. Finally, the outputs from all directions are fused to produce the final feature representation.

transition dynamics for each scanning direction, enabling the model to better capture heterogeneous temporal variations in bitemporal data. Specifically, a learnable drift probability map is estimated from the feature representation

$$v_{k,i} = \sigma\left(f_{\text{PW}}\left(\text{ReLU}\left(f_{\text{PW}}\left(G_{D1}\right)\right)\right)\right) \tag{4}$$

where f_{PW} denotes 1×1 pointwise convolutions, which operate independently at each spatial position without introducing any local receptive field, consistent with the global modeling objective of the GDSM, and σ is the sigmoid function. The estimated drift map encodes the likelihood of temporal inconsistency at each spatial position and is produced independently for each of the $K = 4$ scanning directions, with all parameters optimized end-to-end through the task loss.

The drift probability is subsequently incorporated into the SSM discretization to adapt the global state transition process. First, the time step Δ is modulated as

$$\tilde{\Delta}_{k,i} = \Delta_{k,i}\left(1 + \alpha_k v_{k,i}\right) \tag{5}$$

where $\alpha_k \in R^D$ is a learnable direction-specific coefficient initialized to zero, ensuring the model degrades to the standard SSM at initialization. This formulation enables the model to dynamically adjust the transition rate according to the estimated temporal inconsistency, allowing faster state updates in regions with significant changes while preserving stable dynamics in homogeneous areas. Second, the state transition matrix $\mathbf{A}$ is adapted to regulate the memory decay

$$\tilde{\mathbf{A}}_k = -\exp\left(\mathbf{A}_{\log,k} - |\beta_k|\bar{v}_k\right) \tag{6}$$

where $\beta_k \in R^{D\times N}$ is a learnable per-direction bias initialized to zero, and $\bar{v}_k = \text{mean}_{B,L}(v_k)$ is the spatially averaged drift intensity for direction k. As $\mathbf{A}_k$ is strictly negative, subtracting the nonnegative term $|\beta_k|\bar{v}_k$ from $\mathbf{A}_{\log,k}$ reduces the magnitude of $\mathbf{A}_k$, thereby slowing the state decay and enabling the SSM to retain longer historical context in regions of genuine spectral change, which helps distinguish true land-cover transitions from transient noise or registration errors.

The modified state recurrence for direction k is therefore formulated as

$$\begin{aligned} h_{\text{type}k}(t) &= \tilde{\mathbf{A}}_k\, h_{\text{type}k}(t-1) + \overline{\mathbf{B}}\, S_{\text{type}k}(t) \\ y_{\text{type}k}(t) &= \mathbf{C}\, h_{\text{type}k}(t) + \mathbf{D}\, S_{\text{type}k}(t) \end{aligned} \tag{7}$$

where $\tilde{\mathbf{A}}_k$ is the drift-modulated state transition matrix, $\overline{\mathbf{B}}$ is the input projection matrix, $\mathbf{C}$ and $\mathbf{D}$ are the output projection matrices, and $h_{\text{type}k}(t)$ is the hidden state. In this process, the selective mechanism dynamically adjusts the state transition according to both the input feature responses and the estimated drift probability, allowing the model to emphasize regions with significant bitemporal changes while suppressing redundant background information. The output features obtained from different scanning routes are first reshaped back to the spatial domain and then fused through element-wise summation to form the output feature map

$$\text{TA-SS2D} = \sum_{k=1}^{4} y_{\text{type}k}. \tag{8}$$

After the TA-SS2D operation, the TA-SS2D branch further performs feature transformation and normalization to obtain the global contextual representation. The TA-SS2D branch can therefore be formulated as

$$\begin{aligned} G_{D1} &= \text{SiLU}\left(f_{\text{conv}}\left(f_{\text{Linear}}\left(\text{LN}\left(F_D\right)\right)\right)\right) \\ G_{D2} &= \text{LN}\left(\text{TA-SS2D}\left(G_{D1}\right)\right) \end{aligned} \tag{9}$$

where G_{D1} denotes the transformed feature serving as the input to the TA-SS2D module, and G_{D2} represents the feature embedding after global dependency modeling. Here, f_{Linear}, f_{conv}, SiLU, and LN correspond to the linear transformation, convolution operation, SiLU activation function, and layer normalization, respectively. Subsequently, the output of the multiplicative interaction branch is expressed as

$$\begin{aligned} G_{D3} &= \text{CBAM}\left(\text{SiLU}\left(f_{\text{Linear}}\left(F_D\right)\right)\right) \\ G_{D4} &= G_{D2} G_{D3} \end{aligned} \tag{10}$$

where CBAM refers to the convolutional block attention module [45], which combines channel and spatial attention mechanisms to improve feature representation. Specifically, CBAM sequentially applies channel attention to model interchannel relationships and spatial attention to capture informative regions in the feature map, thereby enhancing the discriminative capability of the extracted features. The feature G_{D3} corresponds to the output of the multiplicative branch, while G_{D4} represents the globally interacted feature obtained after element-wise multiplication. Finally, a residual branch is incorporated to retain the original input information and alleviate potential overfitting. The output of the GDSM can be formulated as

$$G_D = F_D + f_{\text{Linear}}\left(G_{D4}\right) \tag{11}$$

where G_D captures global contextual differential features aggregated along the spatial dimension.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:42:48 UTC from IEEE Xplore. Restrictions apply.

C. Local Differential Sensing Module

To enhance the sensitivity of the network to local structural discrepancies and enable the capture of fine-grained change information along multiple directions, an LDSM based on differential convolutions [46], [47] is introduced. The LDSM further complements the pixel-level global dependency modeling capability of the global processing module by emphasizing localized variations. In our implementation, six convolutional layers are arranged in parallel for feature extraction, consisting of five differential convolution kernels and one conventional convolution kernel. The structures of the differential convolution kernels are illustrated in Fig. 2(c). Unlike conventional convolutions that primarily respond to intensity variations, differential convolutions explicitly embed directional pixel-difference operators into convolution kernels, allowing local gradient priors to be directly incorporated into feature learning. As a result, the network is capable of capturing richer and more fine-grained structural differences. It is worth noting that the convolutional weights in each branch are learnable, while the spatial differential topology is fixed and predefined, thereby constraining feature learning within explicit directional difference patterns rather than allowing unrestricted spatial aggregation. Compared with standard learnable convolutions, this structured design introduces a strong gradient-aware inductive bias that preserves local discontinuity information and improves the separability between changed and unchanged regions.

Specifically, six parallel convolutional branches are employed: a central difference convolution (CDC), a horizontal difference convolution (HDC), a vertical difference convolution (VDC), an anti-diagonal difference convolution (ADC), a main-diagonal difference convolution (MDC), and a standard convolution (SC). The CDC enhances local responses by emphasizing the difference between the central pixel and its surrounding neighbors. The HDC and VDC are designed to extract gradient information along the horizontal and vertical directions, respectively. In contrast, the ADC and MDC extend the differential operation to diagonal orientations, enabling the model to effectively perceive texture variations along oblique directions. The standard convolution branch preserves intensity and texture information, thereby complementing the differential branches in regions where pure difference operators may be insufficient. Since all differential branches operate based on pixel-wise differences, the resulting feature representations exhibit increased sensitivity to gradients, edges, and texture variations. This design significantly improves the model's capability to characterize subtle local structural changes.

To avoid the excessive computational overhead introduced by multiple parallel convolutional branches, we exploit the additivity property of convolution operations to perform kernel reparameterization. Specifically, the convolution kernels from all directional branches are summed along the spatial dimension to construct an equivalent single convolution kernel, thereby enabling efficient parameter reparameterization. The reparameterization process is visually illustrated in Fig. 4. During the backward propagation stage, the kernel parameters of the six parallel convolution branches are updated independently via gradient backpropagation. During the forward propagation stage, the parallel kernels remain fixed and are combined at corresponding spatial locations to compute the reparameterized kernel. Notably, this reparameterization strategy accelerates both training and inference, as it is applied during the forward pass. For clarity, the bias term is omitted, and the reparameterization process is formulated as

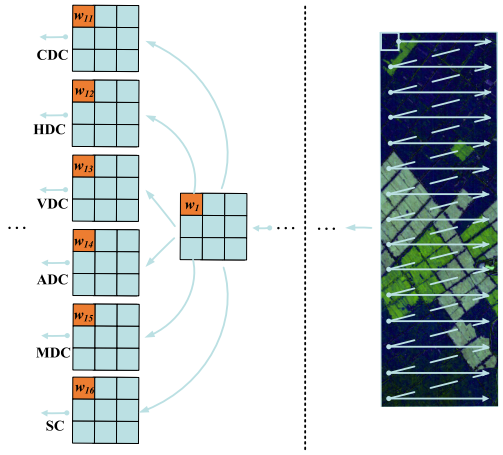


Fig. 4. Kernel reparameterization process of LDSM. CDC, HDC, VDC, ADC, MDC, and SC are parameterized into structured 3 × 3 kernels and fused into a unified kernel w. The fused kernel scans the input feature map, and gradients are backpropagated to update the parameters of all six convolution branches.

$$K_{cvt} = \sum_{i=1}^{6} w_i$$
$$L_D = F_D * K_{cvt} \tag{12}$$

where w_i denotes the reparameterized convolution kernels obtained from the local differential branches, and * represents the convolution operation. L_D denotes the local contextual differential feature extracted along the spatial dimension. The same processing strategy is applied to the concatenated feature F_C to obtain the corresponding local feature representation L_C, which serves as the local input for subsequent feature fusion stages.

D. Feature Fusion Module

To effectively integrate the global and local change features derived from the GDSM and LDSM, an FFM is designed based on differential enhancement and adaptive global–local attention, as shown in Fig. 2(d). The FFM explicitly models the differences between the two feature branches to emphasize the most discriminative change information across spectral–spatial dimensions. Meanwhile, global contextual cues and local differential responses are jointly exploited to generate adaptive fusion weights, allowing the network to more effectively highlight salient change regions during feature integration. Through this selective and adaptive fusion strategy, the FFM maintains global semantic coherence while preserving fine-grained local structural details, thereby enhancing the representational capacity of change features and yielding more reliable and discriminative embeddings for subsequent change identification.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:42:48 UTC from IEEE Xplore. Restrictions apply.

Given the global and local features produced by the two branches, denoted as $G_D \in R^{C_1 \times H \times W}$ and $L_C \in R^{C_1 \times H \times W}$, respectively, the fusion process can be formulated as

$$F_1 = f_{\text{Conv2}}\left(\delta\left(f_{\text{Conv1}}\left(\text{GAP}\left(G_D - L_C\right)\right)\right)\right)$$
$$F_2 = f_{\text{Conv2}}\left(\delta\left(f_{\text{Conv1}}\left(G_D - L_C\right)\right)\right)$$
$$F_3 = f_{\text{Conv2}}\left(\delta\left(f_{\text{Conv1}}\left(\text{GMP}\left(G_D - L_C\right)\right)\right)\right) \quad (13)$$

where GAP and GMP represent global average pooling and global max pooling operations, respectively, f_{Conv1} compresses the channel dimension to C_1/r, δ denotes the ReLU activation, and f_{Conv2} restores the channel dimension to its original size, where r is the channel reduction ratio. The resulting feature representations $\{F_1, F_2, F_3\}$ are then aggregated to generate attention weights for subsequent feature modulation

$$W = \sigma(F_1 + F_2 + F_3)$$
$$F_{\text{out1}} = G_D \odot W + L_C \odot (1 - W) \quad (14)$$

where $\odot$ indicates element-wise multiplication. The three branches have different spatial scales but are broadcast to align with the dimensions of the input features during fusion.

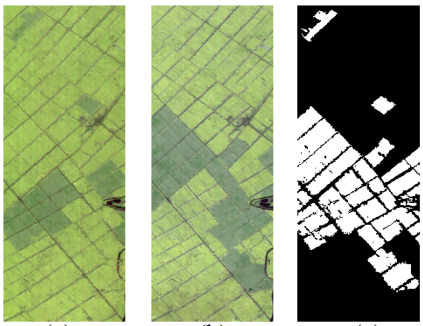
Through the carefully designed differential enhancement mechanism, global semantic modeling, and adaptive weight fusion strategy, the FFM achieves complementary integration of global and local features. The same fusion operation is applied to the global feature from the concatenation branch (G_C) and the local feature from the difference branch (L_D) to obtain F_{out2}. The resulting F_{out2} is then concatenated with F_{out1} for final classification.

E. Training and Loss Function

The proposed method is an image-level CD approach, in which both training and inference are performed on entire bitemporal images. This design enables the model to capture global contextual differences across the full spatial extent. Unlike patch-based methods, the training process is conducted under a segmentation paradigm. During training, a fixed proportion of labeled pixels is randomly sampled from each category to form the training set, and a corresponding training mask is generated simultaneously. Pixels that are not selected for training are excluded from the loss computation through the masking mechanism. Consequently, a masked cross-entropy loss is employed for network optimization, defined as

$$L = -\sum_{h=1}^{H}\sum_{w=1}^{W} m^{(h,w)} \sum_{c=1}^{C} Y_c^{(h,w)} \log\left(\widehat{Y}_c^{(h,w)}\right) \quad (15)$$

where Y denotes the one-hot encoded ground-truth labels, and $\widehat{Y}$ represents the predicted probability map produced by the network for class c. The mask $m \in R^{H \times W}$ indicates the training pixels, with a value of 1 assigned to sampled pixels and 0 to all remaining pixels. Through the masking strategy, only the randomly selected training pixels contribute to the loss computation, while the remaining pixels do not affect parameter updates. This mechanism allows the proposed image-level framework to be effectively trained with limited labeled samples, while preserving global contextual modeling capability.



(a) (b) (c)

Fig. 5. Farmland dataset. (a) HSI was acquired on May 3, 2006. (b) HSI was acquired on April 23, 2007. (c) Ground truth.

F. Patch-Driven GLDMamba-P

For a fair comparison with most existing patchwise methods, a patch-driven variant, termed GLDMamba-P, is constructed based on the same architecture as GLDMamba. In this variant, the patch size and batch size are empirically set to 5×5 and 64 [48], respectively, while other settings are kept consistent with those of GLDMamba. Different from the full-image input manner, GLDMamba-P takes bitemporal HSI patch pairs as input and predicts the class of the central pixel by extracting local contextual differential features from the two temporal patches. In this way, GLDMamba-P preserves the core global–local differential modeling mechanism of GLDMamba while adapting it to a patchwise prediction framework, making it suitable for comparison with conventional patch-based HSI-CD methods.

IV. Experiments

In this section, the proposed GLDMamba framework is comprehensively assessed on three benchmark HSI-CD datasets to demonstrate its effectiveness and robustness. First, the employed datasets are described in detail. Subsequently, the experimental configurations and quantitative evaluation criteria are specified. Then, branch-level and component-level ablation studies are conducted to investigate the contributions of individual modules within the proposed framework. Finally, extensive comparative experiments are conducted to evaluate GLDMamba against several state-of-the-art approaches through both quantitative metrics and qualitative visual comparisons.

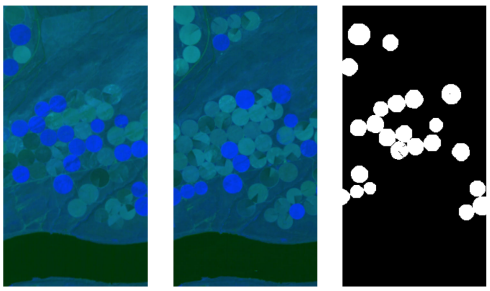
A. Experimental Datasets

The proposed GLDMamba framework is evaluated on three benchmark HSI-CD datasets, namely Farmland, Hermiston, and River, to comprehensively verify its effectiveness. Table I summarizes the detailed class distributions of these datasets.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:42:48 UTC from IEEE Xplore. Restrictions apply.

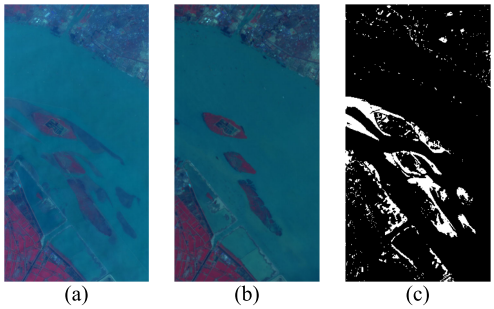
TABLE I
DETAILED CATEGORY INFORMATION FOR THE THREE DATASETS

Dataset	Training Set (1%)			Validation set (1%)			Test set (98%)		
	Unchanged	Change	Total	Unchanged	Change	Total	Unchanged	Change	Total
Farmland	448	183	631	448	183	631	43827	18911	61738
Hermiston	681	100	781	681	100	781	66652	9786	76438
River	1019	97	1116	1019	97	1116	99847	9504	109351



(a) (b) (c)

Fig. 6. Hermiston dataset. (a) HSI was acquired on May 1, 2004. (b) HSI was acquired on May 8, 2007. (c) Ground truth.



(a) (b) (c)

Fig. 7. River dataset. (a) HSI was acquired on May 3, 2013. (b) HSI was acquired on December 31, 2013. (c) Ground truth.

1) Farmland Dataset: The Farmland dataset consists of a pair of multitemporal HSIs and the corresponding change reference, as shown in Fig. 5. The data were acquired by the EO-1 hyperspectral sensor during 2006–2007 over agricultural areas near Yancheng City, China. Each image has a spatial size of 450 × 140 pixels with 155 spectral bands.

2) Hermiston Dataset: Fig. 6 illustrates the Hermiston dataset, which includes two HSIs captured by the EO-1 sensor in 2004 and 2007 and the associated change annotations. The images have a spatial resolution of 390 × 200 pixels and consist of 242 spectral bands.

3) River Dataset: The River dataset, shown in Fig. 7, comprises two EO-1 HSIs acquired on May 3 and December 31, 2013, over a watershed area in Jiangsu Province, China. The data cube has a spatial dimension of 463 × 241 pixels with 198 spectral bands covering 400–2500 nm.

B. Experiment Setup

1) Evaluation Criteria: To comprehensively and quantitatively evaluate the CD performance of different methods, five metrics are adopted in this article r: overall accuracy (OA), Kappa coefficient (Kappa), precision (P), recall (R), and $F1$-score ($F1$). Their definitions are given as follows:

$$\begin{aligned} \mathrm{OA} &= \frac{\mathrm{TP}+\mathrm{TN}}{\mathrm{TP}+\mathrm{TN}+\mathrm{FP}+\mathrm{FN}}, \quad \mathrm{Kappa} = \frac{\mathrm{OA}-P_e}{1-P_e} \\ \mathrm{F1} &= \frac{2\mathrm{P}\times\mathrm{R}}{\mathrm{P}+\mathrm{R}}, \quad \mathrm{P} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}}, \quad \mathrm{R} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}} \\ P_e &= \frac{(\mathrm{TP}+\mathrm{FP})(\mathrm{TP}+\mathrm{FN})+(\mathrm{TN}+\mathrm{FN})(\mathrm{TN}+\mathrm{FP})}{(\mathrm{TP}+\mathrm{TN}+\mathrm{FP}+\mathrm{FN})^2} \end{aligned} \tag{16}$$

where TP denotes the number of pixels that are labeled as changed and correctly detected as changed, FP denotes the number of unchanged pixels that are incorrectly detected as changed, TN denotes the number of unchanged pixels that are correctly identified as unchanged, and FN denotes the number of changed pixels that are incorrectly detected as unchanged.

2) Implementation Details: All experiments are implemented within the PyTorch framework and performed on a workstation equipped with an NVIDIA RTX 3090 GPU with 24 GB memory. The proposed GLDMamba is optimized using Adam with a learning rate of 0.0001, and the network is optimized for 300 training epochs, and the convergence behavior is monitored based on validation accuracy. A StepLR learning rate scheduler is employed to gradually decrease the learning rate with a step size of 30 and a decay factor of 0.9. To ensure fair and stable comparisons, all reported results are presented as the mean and standard deviation over ten runs with different random seeds. For all methods, 1% of the labeled samples are randomly selected for training, 1% for validation, and the remaining 98% are used for testing. When benchmarking against existing methods, we strictly follow their original implementations with clearly specified parameters and training procedures. For methods lacking sufficient implementation details, the same training protocol as that of the proposed method is adopted to ensure consistency.

3) Comparison Methods: Nine representative and advanced DL-based HSI-CD methods are selected for comparison, including CNN and graph convolution-based methods SRGMGN [49], Siamese and multiscale CNN-based methods CSANet [50] and GTMSiam [51], transformer-based methods DBS3TAN [18], SSTFormer [42] and GLAFormer [43], Mamba-based methods SSMIF [27] and GASSM [28], and entropy-driven attention-based method AIWSEN [52].

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:42:48 UTC from IEEE Xplore. Restrictions apply.

TABLE II
CD RESULTS ON THE FARMLAND, HERMISTON, AND RIVER DATASETS WITH 1% LABELED SAMPLES

Dataset	Method	OA(%)	Kappa(×100)	F1(%)	P(%)	R(%)
Farmland	DBS3TAN	94.91 ± 1.06	87.67 ± 2.55	91.25 ± 1.82	91.27 ± 3.98	91.55 ± 4.40
	SRGMGN	95.38 ± 0.52	88.57 ± 1.42	91.77 ± 1.09	94.75 ± 1.41	89.07 ± 3.07
	CSANet	96.28 ± 0.22	91.02 ± 0.54	93.65 ± 0.39	92.90 ± 1.18	94.44 ± 1.33
	SSTFormer	96.42 ± 0.31	91.34 ± 0.93	93.87 ± 0.65	93.25 ± 1.33	94.53 ± 1.09
	GTMSiam	97.13 ± 0.23	93.02 ± 0.55	95.05 ± 0.39	95.10 ± 0.74	95.00 ± 0.57
	GLAFormer	96.46 ± 0.33	91.44 ± 0.78	93.93 ± 0.54	93.58 ± 1.63	94.34 ± 1.50
	SSMIF	96.40 ± 0.36	91.34 ± 0.81	93.89 ± 0.55	92.69 ± 1.51	95.14 ± 0.69
	GASSM	97.05 ± 0.16	92.86 ± 0.40	94.93 ± 0.29	94.74 ± 0.97	95.15 ± 1.13
	AIWSEN	97.30 ± 0.21	93.50 ± 0.48	95.41 ± 0.33	94.11 ± 1.14	96.79 ± 0.81
	GLDMamba-P	97.53 ± 0.10	94.02 ± 0.25	95.75 ± 0.19	95.72 ± 0.73	95.79 ± 0.87
	GLDMamba	**97.95 ± 0.16**	**95.05 ± 0.38**	**96.50 ± 0.28**	**95.76 ± 0.40**	**97.25 ± 0.71**
Hermiston	DBS3TAN	94.14 ± 0.43	68.34 ± 2.05	71.37 ± 1.84	95.71 ± 4.81	57.01 ± 2.15
	SRGMGN	94.59 ± 1.13	73.58 ± 7.84	76.56 ± 7.42	84.55 ± 3.47	71.23 ± 11.84
	CSANet	98.53 ± 0.13	93.50 ± 0.54	94.34 ± 0.46	93.36 ± 1.77	95.40 ± 1.44
	SSTFormer	98.55 ± 0.11	93.60 ± 0.99	94.43 ± 0.77	93.18 ± 2.50	95.81 ± 1.57
	GTMSiam	98.89 ± 0.06	95.02 ± 0.26	95.66 ± 0.23	95.93 ± 0.76	95.41 ± 0.87
	GLAFormer	98.72 ± 0.10	94.29 ± 0.46	95.03 ± 0.40	94.64 ± 1.41	95.46 ± 1.50
	SSMIF	98.68 ± 0.15	94.08 ± 0.67	94.84 ± 0.59	94.65 ± 0.80	95.04 ± 1.19
	GASSM	98.69 ± 0.07	94.14 ± 0.35	94.89 ± 0.30	95.11 ± 0.67	94.67 ± 0.87
	AIWSEN	98.84 ± 0.12	94.81 ± 0.52	95.48 ± 0.45	94.92 ± 1.10	96.06 ± 0.64
	GLDMamba-P	98.84 ± 0.09	94.83 ± 0.41	95.50 ± 0.36	94.96 ± 1.25	**96.07 ± 1.31**
	GLDMamba	**98.94 ± 0.04**	**95.23 ± 0.20**	**95.84 ± 0.17**	**96.13 ± 0.72**	95.56 ± 0.75
River	DBS3TAN	93.25 ± 0.83	39.68 ± 16.42	42.12 ± 17.19	83.75 ± 11.58	31.81 ± 16.72
	SRGMGN	93.86 ± 0.94	46.88 ± 15.51	49.34 ± 15.89	84.68 ± 6.48	37.70 ± 16.32
	CSANet	95.42 ± 0.13	66.86 ± 1.13	69.39 ± 1.05	79.11 ± 1.91	61.90 ± 2.15
	SSTFormer	95.16 ± 0.23	65.96 ± 2.87	68.53 ± 2.76	78.80 ± 3.52	60.85 ± 4.19
	GTMSiam	96.59 ± 0.21	77.39 ± 1.58	79.23 ± 1.48	84.20 ± 1.75	74.90 ± 2.73
	GLAFormer	95.30 ± 0.24	66.60 ± 2.94	69.07 ± 2.89	80.80 ± 2.86	60.68 ± 5.28
	SSMIF	95.26 ± 0.14	66.89 ± 1.14	69.41 ± 1.09	79.10 ± 1.94	61.92 ± 2.13
	GASSM	96.48 ± 0.19	75.44 ± 2.12	77.32 ± 2.07	**87.85 ± 3.24**	69.37 ± 4.73
	AIWSEN	96.39 ± 0.17	75.80 ± 1.45	77.76 ± 1.37	83.66 ± 1.54	72.72 ± 2.76
	GLDMamba-P	96.81 ± 0.10	**79.43 ± 0.56**	**81.17 ± 0.52**	83.45 ± 1.89	**79.10 ± 1.97**
	GLDMamba	**96.86 ± 0.18**	79.17 ± 1.55	80.87 ± 1.47	85.49 ± 1.91	76.86 ± 3.37

C. Experimental Results

Table II presents the CD results on three benchmark datasets using only 1% of the ground-truth samples for training, where the best results are highlighted in bold and the second-best are underlined. Considering the class imbalance caused by the relatively small proportion of changed areas, OA and the Kappa are jointly employed to evaluate overall performance. Meanwhile, the $F1$, defined as the harmonic mean of P and R, is adopted for comprehensive assessment, while P and R are further analyzed individually to provide more fine-grained insights into model behavior.

On the Farmland dataset, the proposed GLDMamba achieves the best performance across all five evaluation metrics, with an OA of 97.95%, a Kappa coefficient of 0.9505, and an $F1$-score of 96.50%. Compared with the Mamba-based SSMIF and GASSM, its $F1$-score is significantly improved. This improvement indicates that applying standard Mamba within a patch-based framework is insufficient to fully exploit long-range dependencies in HSI-CD. In contrast, the proposed GDSM employs the TA-SS2D mechanism, which adaptively adjusts the state transition process according to bitemporal difference responses and aggregates contextual information through multidirectional scanning, thereby enabling more effective full-scene global dependency modeling. Meanwhile, GLDMamba directly processes image-level bitemporal HSIs, avoiding the receptive-field limitation caused by local patch inputs. GLDMamba-P also outperforms other patch-based methods, demonstrating that the proposed global–local differential design remains effective even under the patch-based setting. In addition, AIWSEN shows competitive performance due to its entropy-driven feature enhancement and bitemporal interaction modeling. Fig. 8 shows the change detection maps generated by different methods on the Farmland dataset, where white denotes TP, black denotes TN, red corresponds to FP, and blue denotes FN. Compared with other patch-based DL methods, GLDMamba and GLDMamba-P produce

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:42:48 UTC from IEEE Xplore. Restrictions apply.

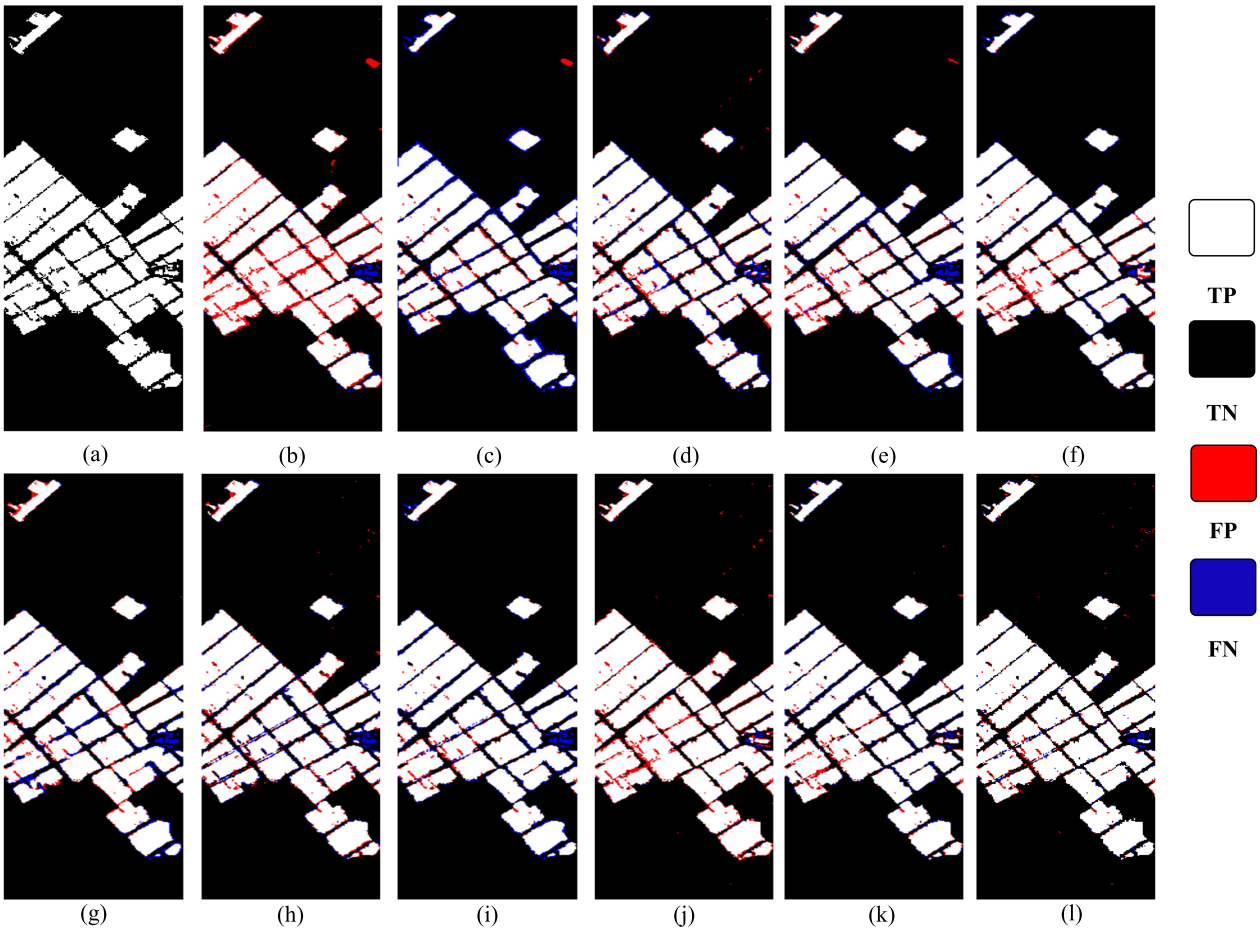


Fig. 8. CD results on the Farmland dataset. (a) GT. (b) DBS3TAN. (c) SRGMGN. (d) CSANet. (e) SSTFormer. (f) GTMSiam. (g) GLAFormer. (h) SSMIF. (i) GASSM. (j) AIWSEN. (k) GLDMamba-P. (l) GLDMamba.

more continuous change regions, clearer boundaries, and fewer scattered false alarms and missed detections. These results verify that the proposed image-level global–local differential framework can effectively capture subtle farmland changes while suppressing background interference.

On the Hermiston dataset, the proposed GLDMamba achieves the best overall performance across most evaluation metrics, including an OA of 98.94%, a Kappa coefficient of 0.9523, an $F1$-score of 95.84%, and a precision of 96.13%. Compared with the Farmland and River datasets, the Hermiston dataset contains more distinguishable change patterns, allowing recent DL methods to achieve relatively high performance. Among them, GTMSiam obtains the second-best overall performance, indicating that its Siamese multiscale convolutional structure can effectively extract local bitemporal difference features when changed and unchanged regions are relatively easy to separate. However, GLDMamba still achieves a moderate but stable improvement over GTMSiam, mainly benefiting from the LDSM. By introducing multiple directional differential convolution kernels, the LDSM explicitly enhances local gradient, edge, and structural difference responses, which helps the model better preserve fine-grained boundaries and detect subtle local changes. In addition, the GDSM provides full-scene contextual constraints to maintain spatial consistency, while the FFM adaptively integrates global and local features to generate more discriminative change representations. Notably, although AIWSEN and GLDMamba-P obtain slightly higher recall, GLDMamba maintains a better balance between precision and recall, thereby achieving the highest $F1$-score. This demonstrates its robustness and effectiveness in accurately capturing change information while suppressing false detections on the Hermiston dataset. Fig. 9 shows the CD maps generated by each method on this dataset. It can be observed that GLDMamba produces noticeably clearer results with more detailed and well-defined boundaries. In addition, it successfully identifies most change regions that are missed by other methods. These observations highlight the advantage of GLDMamba in capturing subtle and small-scale changes with high spatial consistency.

On the River dataset, GLDMamba achieves an OA of 96.86% and a higher precision than GLDMamba-P, while its Kappa value and $F1$-score rank second among all methods. In contrast, GLDMamba-P obtains the highest Kappa value, $F1$-score, and recall. Therefore, the River results do not imply the absolute superiority of image-level global modeling over patch-based local modeling; instead, they reveal a clear

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:42:48 UTC from IEEE Xplore. Restrictions apply.

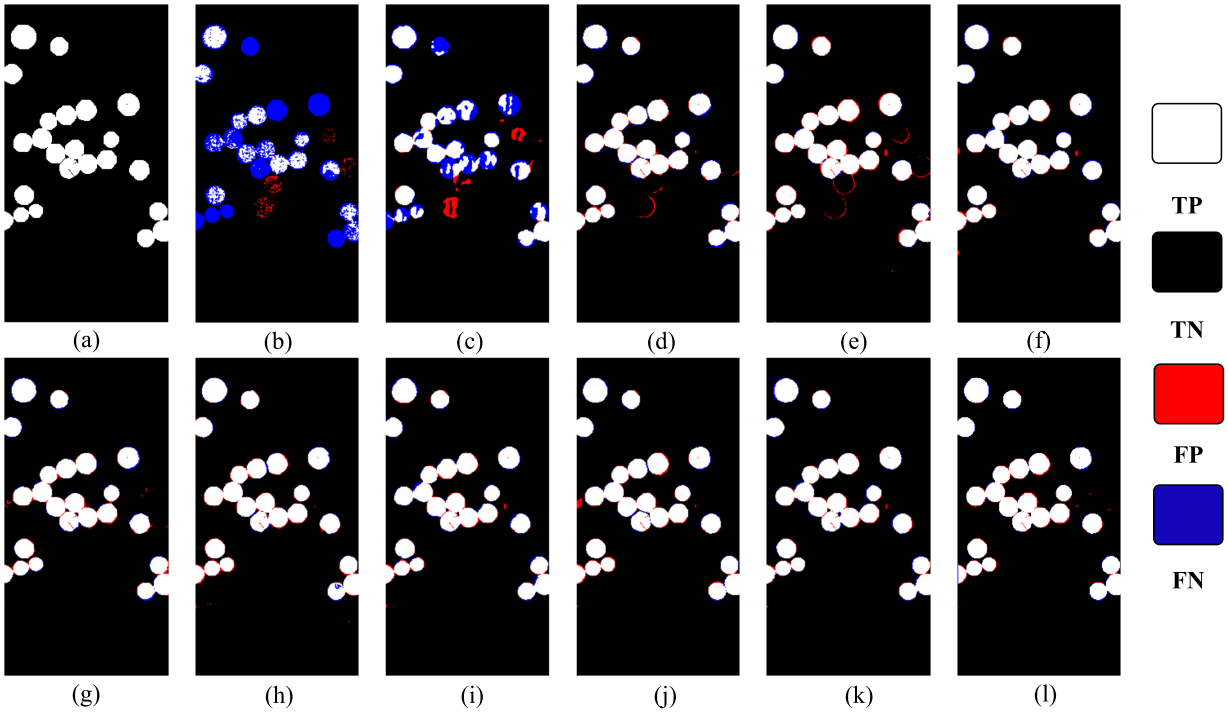


Fig. 9. CD results on the Hermiston dataset. (a) GT. (b) DBS3TAN. (c) SRGMGN. (d) CSANet. (e) SSTFormer. (f) GTMSiam. (g) GLAFormer. (h) SSMIF. (i) GASSM. (j) AIWSEN. (k) GLDMamba-P. (l) GLDMamba.

precision–recall tradeoff between the two strategies. The River dataset contains sparse, fragmented, and spatially irregular changed regions, where local patch-based modeling can better capture neighborhood-level spectral–spatial variations and thus improve recall, but may also introduce more false positives. By comparison, GLDMamba benefits from full-image contextual constraints through image-level processing, which enhances spatial consistency, suppresses isolated noisy responses, and improves precision. GASSM achieves the highest precision but has a relatively lower recall, suggesting a tendency toward missed detections, whereas GLDMamba provides a relatively balanced result by combining global dependency modeling with local differential perception. Fig. 10 illustrates the visual CD results on the River dataset using various methods. Earlier methods, such as DBS3TAN and SRGMGN, produce significant noise and missed detections, as evidenced by the scattered blue and red regions. In contrast, GLDMamba generates more coherent spatiotemporal change regions with clearer boundaries and fewer false alarms. These observations indicate that GLDMamba is effective in suppressing noise and maintaining global spatial consistency, whereas GLDMamba-P is advantageous when the recall of sparse and fragmented changed regions is prioritized.

Across the three benchmark datasets, GLDMamba achieves the best OA and generally competitive or superior Kappa and $F1$ performance. The high OA and Kappa values indicate its effectiveness in feature extraction and its ability to alleviate class imbalance between changed and unchanged regions, while the strong $F1$ reflects a favorable tradeoff between false alarms and missed detections. These results suggest that GLDMamba is sensitive to subtle changes and can effectively suppress interference from unchanged areas. Moreover, the comparison between GLDMamba and its patch-based variant GLDMamba-P shows that image-level global modeling and patch-based local modeling exhibit different precision–recall tradeoffs. GLDMamba generally provides stronger spatial coherence and fewer false alarms, while GLDMamba-P can achieve higher recall in sparse and fragmented change scenarios such as the River dataset. Overall, this confirms the effectiveness of the proposed architecture in jointly exploiting global and local information for HSI-CD.

D. Discussion

1) Ablation Studies: To comprehensively evaluate the effectiveness of the proposed GLDMamba, extensive ablation experiments are conducted from both branch-level and component-level perspectives, with the corresponding quantitative results reported in Tables III and IV. These experiments aim to systematically analyze the contribution of the dual-branch interaction mechanism and the key functional modules within the framework.

At the branch level, the results demonstrate that the dual-branch cross-processing strategy significantly outperforms single-branch configurations. By enabling more effective interaction between bitemporal features, this design enhances the representation of change information and leads to improved classification accuracy. In addition, the impact of the absolute operation in difference computation is further investigated. The results show that replacing the absolute difference with signed difference causes a clear performance decline, indicating that the absolute operation plays a crucial role in stabilizing feature representation and improving detection reliability.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:42:48 UTC from IEEE Xplore. Restrictions apply.

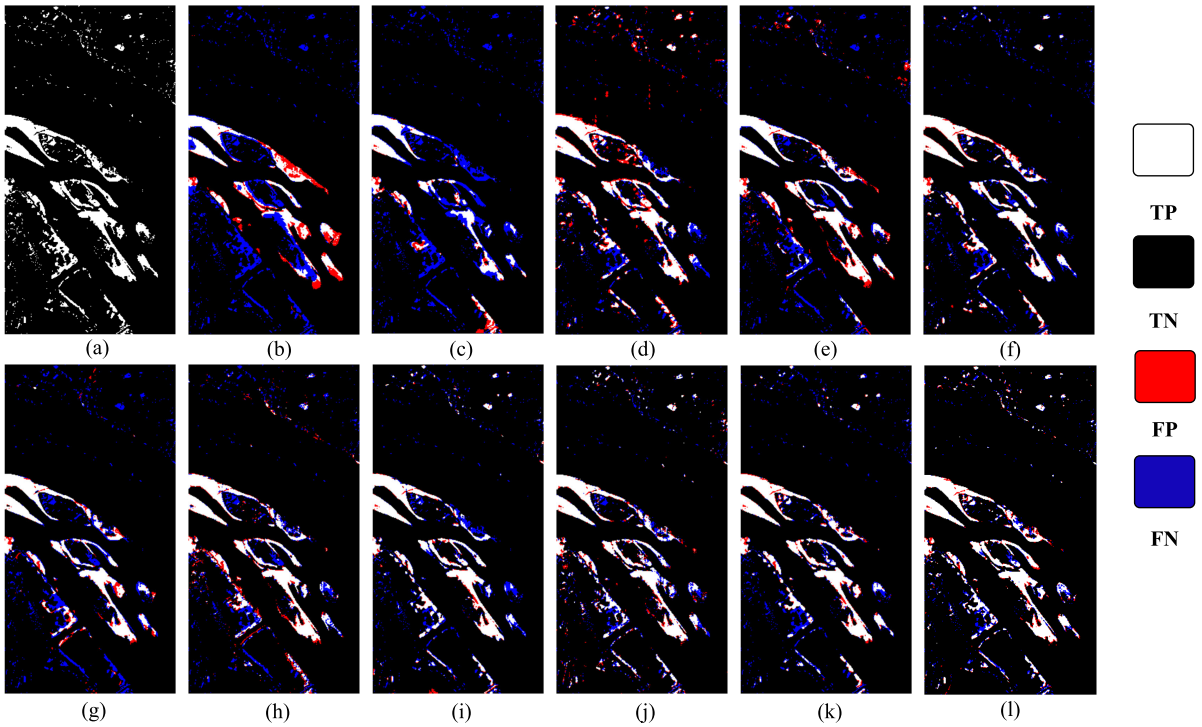


Fig. 10. CD results on the River dataset. (a) GT. (b) DBS3TAN. (c) SRGMGN. (d) CSANet. (e) SSTFormer. (f) GTMSiam. (g) GLAFormer. (h) SSMIF. (i) GASSM. (j) AIWSEN. (k) GLDMamba-P. (l) GLDMamba.

At the component level, ablation results reveal that removing any of the core modules, namely GDSM, LDSM, or FFM, leads to notable performance degradation. Specifically, GDSM contributes to capturing large-scale consistent changes and modeling long-range dependencies through global differential learning, thereby enhancing robustness. LDSM improves sensitivity to fine-grained variations, such as boundary details and small targets, enabling more precise localization of change regions. Meanwhile, FFM effectively fuses global and local differential features within deep semantic spaces, suppressing redundant and noisy responses while reducing false detections. Collectively, these modules work in a complementary manner, ensuring the overall effectiveness and stability of the proposed framework.

2) Component Substitution Analysis: To further evaluate the impact of individual components within the GLDMamba framework, we conducted a series of experiments by substituting different components of the full model. Specifically, we examined the effects of replacing the differential convolution (DC) with an SC, as well as substituting the TA-SS2D module with the SS2D module. The performance results, as shown in Table V, demonstrate the effectiveness of the full model. In all three datasets, the full GLDMamba model consistently outperformed the component substitution configurations in terms of OA, Kappa, and $F1$. This indicates that the integration of the various components in the full model is crucial for capturing more complex spatial and temporal features, thereby enhancing the model's performance.

3) Directional Adaptability of TA-SS2D: To evaluate whether the four-direction scanning strategy in TA-SS2D introduces directional bias, a robustness experiment is conducted on synthetic bitemporal image pairs. As illustrated in Fig. 11, five composite datasets with added background noise and change targets of different orientations are constructed, including horizontal, vertical, 45° diagonal, 135° diagonal, and circular patterns, each with a resolution of 256 × 256. The model based on TA-SS2D is trained using only 5% labeled pixels and evaluated on these datasets, with

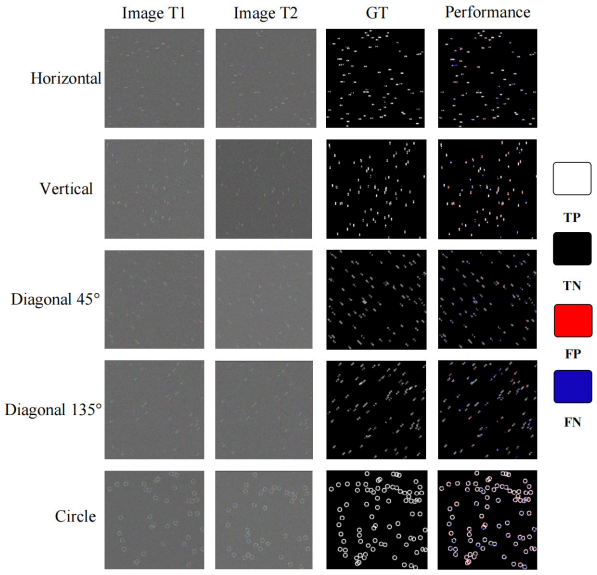


Fig. 11. CD results on synthetic datasets with five change orientations.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:42:48 UTC from IEEE Xplore. Restrictions apply.

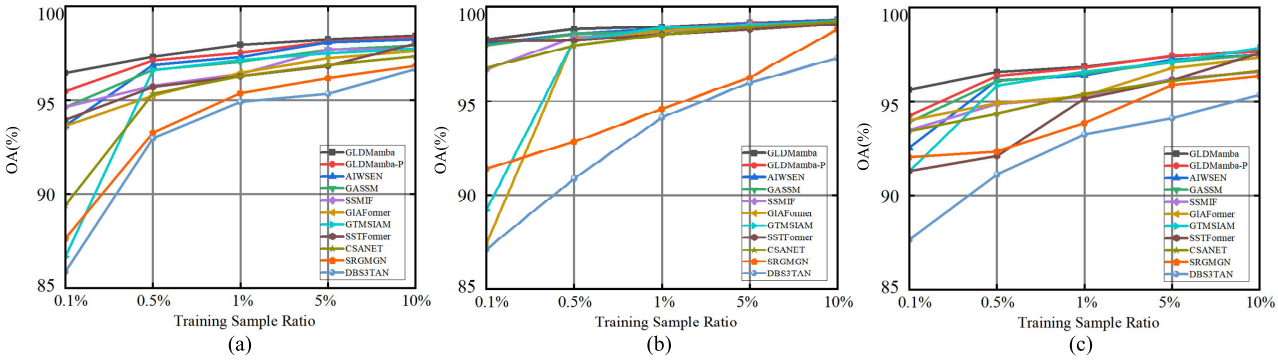


Fig. 12. CD performance of different methods under different training sample ratios. (a) Farmland. (b) Hermiston. (c) River.

TABLE III

BRANCHING ABLATION EXPERIMENTS FOR THREE DATASETS

Dataset	Configuration	OA(%)	Kappa(×100)	F1(%)
Farmland	w/o Abs	97.41 ± 0.32	93.56 ± 0.76	95.48 ± 0.51
	w/o Concat	97.33 ± 0.14	93.54 ± 0.35	95.42 ± 0.25
	w/o Diff	96.75 ± 0.10	92.14 ± 0.24	94.43 ± 0.17
	Full model	**97.95 ± 0.16**	**95.05 ± 0.38**	**96.50 ± 0.28**
Hermiston	w/o Abs	98.84 ± 0.08	94.80 ± 0.38	95.47 ± 0.33
	w/o Concat	98.53 ± 0.14	93.47 ± 0.62	94.31 ± 0.54
	w/o Diff	98.79 ± 0.13	94.59 ± 0.59	95.29 ± 0.51
	Full model	**98.94 ± 0.04**	**95.23 ± 0.20**	**95.84 ± 0.17**
River	w/o Abs	96.38 ± 0.29	72.35 ± 2.38	74.56 ± 2.22
	w/o Concat	96.39 ± 0.16	77.07 ± 1.12	79.04 ± 1.04
	w/o Diff	94.48 ± 0.37	63.05 ± 1.91	66.03 ± 1.80
	Full model	**96.86 ± 0.18**	**79.17 ± 1.55**	**80.87 ± 1.47**

TABLE IV

COMPONENT ABLATION EXPERIMENTS OF THE THREE DATASETS

Dataset	Configuration	OA(%)	Kappa(×100)	F1(%)
Farmland	w/o GDSM	97.39 ± 0.15	93.69 ± 0.37	95.52 ± 0.27
	w/o LDSM	97.50 ± 0.13	93.95 ± 0.32	95.70 ± 0.22
	w/o FFM	97.83 ± 0.14	94.73 ± 0.33	96.27 ± 0.24
	Full model	**97.95 ± 0.16**	**95.05 ± 0.38**	**96.50 ± 0.28**
Hermiston	w/o GDSM	98.78 ± 0.11	94.51 ± 0.51	95.20 ± 0.44
	w/o LDSM	98.86 ± 0.05	94.92 ± 0.21	95.57 ± 0.18
	w/o FFM	98.91 ± 0.10	95.14 ± 0.41	95.77 ± 0.36
	Full model	**98.94 ± 0.04**	**95.23 ± 0.20**	**95.84 ± 0.17**
River	w/o GDSM	95.93 ± 0.17	70.83 ± 1.46	72.97 ± 1.39
	w/o LDSM	96.54 ± 0.24	78.07 ± 1.79	79.96 ± 1.67
	w/o FFM	96.75 ± 0.16	79.08 ± 1.02	80.72 ± 0.95
	Full model	**96.86 ± 0.18**	**79.17 ± 1.55**	**80.87 ± 1.47**

TABLE V

PERFORMANCE COMPARISON UNDER COMPONENT SUBSTITUTION WITHIN THE FULL GLDMAMBA FRAMEWORK

Dataset	Configuration	OA(%)	Kappa(×100)	F1(%)
Farmland	DC→SC	97.67 ± 0.27	94.38 ± 0.65	96.02 ± 0.46
	TA-SS2D→SS2D	97.72 ± 0.23	94.50 ± 0.54	96.12 ± 0.38
	Full model	**97.95 ± 0.16**	**95.05 ± 0.38**	**96.50 ± 0.28**
Hermiston	DC→SC	98.88 ± 0.08	95.00 ± 0.37	95.65 ± 0.33
	TA-SS2D→SS2D	98.84 ± 0.08	94.79 ± 0.36	95.46 ± 0.31
	Full model	**98.94 ± 0.04**	**95.23 ± 0.20**	**95.84 ± 0.17**
River	DC→SC	96.65 ± 0.11	77.74 ± 1.22	79.56 ± 1.18
	TA-SS2D→SS2D	96.73 ± 0.15	78.52 ± 1.06	80.30 ± 0.99
	Full model	**96.86 ± 0.18**	**79.17 ± 1.55**	**80.87 ± 1.47**

the quantitative results reported in Table VI. The results demonstrate that TA-SS2D achieves consistently stable and uniform performance across all directional settings, indicating strong generalization capability to arbitrarily oriented changes.

TABLE VI

CD RESULTS ON SYNTHETIC DATASETS WITH FIVE CHANGE ORIENTATIONS

Orientation	OA(%)	Kappa(×100)	F1(%)
Horizontal	91.53±1.75	84.25±1.70	84.54±1.67
Vertical	93.09±1.40	85.32±1.92	85.60±1.89
Diagonal 45°	91.75±1.81	86.47±1.57	86.72±1.55
Diagonal 135°	92.03±1.32	87.02±1.58	87.25±1.55
Circle	93.36±0.93	86.30±1.10	87.31±1.04

This robustness is attributed to the complementary design of multidirectional scanning, which ensures comprehensive spatial coverage, as well as the drift-aware discretization mechanism, which adaptively enhances responses to changed pixels regardless of their orientation.

4) Impact of Training Sample Ratios: Fig. 12 depicts the CD performance of different methods on the three benchmark datasets under varying training sample ratios. The results indicate that GLDMamba consistently surpasses competing approaches across all training proportions. As the amount of training data increases, the performance gap among different models gradually narrows, indicating that sufficient labeled data significantly enhances feature learning capability for change regions. However, under extremely limited training conditions (e.g., a sampling ratio of 0.1%), GLDMamba exhibits a pronounced advantage over other methods. This observation highlights the critical benefit of leveraging global image-level information to maintain reliable performance in data-scarce scenarios.

5) Computational Complexity Analysis: Table VII presents a comparison of parameter counts (Params) and floating-point operations (FLOPs) for different models on three datasets, providing an intuitive assessment of computational complexity. Leveraging the Mamba architecture's capability to efficiently model long-range dependencies with a compact parameterization, GLDMamba maintains relatively low parameter counts and FLOPs, significantly lower than similar models based on transformer. Despite its high efficiency, GLDMamba still delivers superior accuracy, demonstrating a favorable tradeoff between performance and computational cost.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:42:48 UTC from IEEE Xplore. Restrictions apply.

TABLE VII
COMPARISON OF MODEL COMPLEXITY AMONG DIFFERENT MODELS ON THREE DATASETS

Datasets	Method	DBS3TAN	SRGMGN	CSANet	SSTFormer	GTMSiam	GLAFormer	SSMIF	GASSM	AIWSEN	GLDMamba-P	GLDMamba
Farmland	Params	614.45M	99.92K	40.42M	2.50M	**60.59K**	12.64M	3.43M	716.49K	12.23M	649.51K	649.51K
	FLOPs	59.72G	19.67G	112.47G	730.06G	**1.07G**	1211.52G	50.37G	30.42G	73.98G	6.25G	26.21G
Hermiston	Params	939.48M	122.71K	40.47M	2.55M	**65.81K**	12.69M	3.53M	1.38M	12.79M	666.22K	666.22K
	FLOPs	116.36G	29.89M	146.82G	941.88G	**1.66G**	1504.98G	70.18G	86.21G	182.49G	8.39G	35.64G
River	Params	773.20M	111.19K	40.44M	2.52M	**63.17K**	12.67M	3.48M	1.01M	12.50M	657.77K	657.77K
	FLOPs	138.66G	38.75G	186.47G	1319.11G	**2.14G**	2147.18G	94.62G	77.04G	376.71G	11.52G	48.27G

V. CONCLUSION

This article proposes a novel GLDMamba framework for HSI-CD. The proposed method directly models bitemporal HSIs at the image level, thereby alleviating the inherent limitations of conventional patch-based frameworks. By jointly exploiting a GDSM and an LDSM, GLDMamba captures cross-region change patterns from a holistic perspective while accurately characterizing fine-grained local variations. Furthermore, an FFM is devised to effectively integrate global and local differential features, ensuring semantic consistency and preserving structural details. Extensive experiments on three benchmark HSI-CD datasets demonstrate that GLDMamba consistently outperforms several state-of-the-art methods under varying training sample ratios, while maintaining a relatively low parameter complexity. These results demonstrate the effectiveness and reliability of the proposed framework, particularly in complex scenes and small-sample settings.

REFERENCES

[1] P. Ghamisi et al., "Advances in hyperspectral image and signal processing: A comprehensive overview of the state of the art," *IEEE Geosci. Remote Sens. Mag.*, vol. 5, no. 4, pp. 37–78, Dec. 2017.

[2] N. Farmonov, M. Esmaeili, D. Abbasi-Moghadam, A. Sharifi, K. Amankulova, and L. Mucsi, "HypsLiDNet: 3-D–2-D CNN model and spatial–spectral morphological attention for crop classification with DESIS and LiDAR data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 11969–11996, 2024.

[3] S. Asadzadeh, W. J. D. Oliveira, and C. R. D. Souza Filho, "UAV-based remote sensing for the petroleum industry and environmental monitoring: State-of-the-art and perspectives," *J. Petroleum Sci. Eng.*, vol. 208, Jan. 2022, Art. no. 109633.

[4] Y. Liu, C. Pang, Z. Zhan, X. Zhang, and X. Yang, "Building change detection for remote sensing images using a dual-task constrained deep Siamese convolutional network model," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 5, pp. 811–815, May 2021.

[5] S. Liu, D. Marinelli, L. Bruzzone, and F. Bovolo, "A review of change detection in multitemporal hyperspectral images: Current techniques, applications, and challenges," *IEEE Geosci. Remote Sens. Mag.*, vol. 7, no. 2, pp. 140–158, Jun. 2019.

[6] Z. Zheng, Y. Zhong, J. Wang, A. Ma, and L. Zhang, "Building damage assessment for rapid disaster response with a deep object-based semantic change detection framework: From natural disasters to man-made disasters," *Remote Sens. Environ.*, vol. 265, Nov. 2021, Art. no. 112636.

[7] B. Cui, Y. Zhang, L. Yan, J. Wei, and Q. Huang, "A SAR change detection method based on the consistency of single-pixel difference and neighbourhood difference," *Remote Sens. Lett.*, vol. 10, no. 5, pp. 488–495, 2019.

[8] J. S. Deng, K. Wang, Y. H. Deng, and G. J. Qi, "PCA-based land-use change detection and analysis using multitemporal and multisensor satellite data," *Int. J. Remote Sens.*, vol. 29, no. 16, pp. 4823–4838, 2008.

[9] C. Wu, B. Du, and L. Zhang, "Slow feature analysis for change detection in multispectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 5, pp. 2858–2874, May 2014.

[10] F. Bovolo, L. Bruzzone, and M. Marconcini, "A novel approach to unsupervised change detection based on a semisupervised SVM and a similarity measure," *IEEE Trans. Geosci. Remote Sens.*, vol. 46, no. 7, pp. 2070–2082, Jul. 2008.

[11] Y. Zhan, K. Fu, M. Yan, X. Sun, H. Wang, and X. Qiu, "Change detection based on deep Siamese convolutional network for optical aerial images," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 10, pp. 1845–1849, Oct. 2017.

[12] A. Raza, H. Huo, and T. Fang, "EUNet-CD: Efficient UNet++ for change detection of very high-resolution remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[13] S. Fang, K. Li, J. Shao, and Z. Li, "SNUNet-CD: A densely connected Siamese network for change detection of VHR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[14] C. Zhang et al., "A deeply supervised image fusion network for change detection in high resolution bi-temporal remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 166, pp. 183–200, Aug. 2020.

[15] C. Zhang, L. Wang, S. Cheng, and Y. Li, "SwinSUNet: Pure transformer network for remote sensing image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5224713.

[16] H. Chen, Z. Qi, and Z. Shi, "Remote sensing image change detection with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5607514.

[17] A. Shafique, S. T. Seydi, T. Alipour-Fard, G. Cao, and D. Yang, "SSViT-HCD: A spatial–spectral convolutional vision transformer for hyperspectral change detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 6487–6504, 2023.

[18] Y. Zhang, T. Wang, C. Zhang, S. Xu, H. Gao, and C. Li, "A dual-branch Siamese spatial–spectral transformer attention network for hyperspectral image change detection," *Expert Syst. Appl.*, vol. 238, Mar. 2024, Art. no. 122125.

[19] Z. Chen et al., "TL2GH^{2}T: Triple-path local-to-global network with hybrid head transformer for hyperspectral change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5528112.

[20] A. Gu and T. Dao, "Mamba: Linear-time sequence modeling with selective state spaces," 2023, *arXiv:2312.00752*.

[21] J. Jiao et al., "VMamba: Visual state space model," in *Proc. Adv. Neural Inf. Process. Syst.*, 2024, pp. 103031–103063.

[22] Z. Xie, L. Lv, H. Gao, S. Xu, and H. Xie, "Dual-feature attention hybrid GCN mamba network for joint hyperspectral and LiDAR classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5406514.

[23] L. Lv et al., "Contrastive prototype clustering for multimodal remote sensing data based on spectral–spatial cross mamba," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 19, pp. 8056–8070, 2026.

[24] Y. Li, Y. Luo, L. Zhang, Z. Wang, and B. Du, "MambaHSI: Spatial–spectral mamba for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5524216.

[25] D. Chen, J. Zhang, and J. Li, "UNMamba: Cascaded spatial–spectral mamba for blind hyperspectral unmixing," *IEEE Geosci. Remote Sens. Lett.*, vol. 22, pp. 1–5, 2025.

[26] G. Fu, F. Xiong, J. Lu, and J. Zhou, "SSUMamba: Spatial–spectral selective state space model for hyperspectral image denoising," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5527714.

[27] L. Wu, J. Peng, B. Yang, W. Sun, and Z. Ye, "SSMIF: Enhanced spatial–spectral mamba interactive fusion network for hyperspectral change detection," in *Proc. IEEE Int. Conf. Signal, Inf. Data Process. (ICSIDP)*, Nov. 2024, pp. 1–5.

[28] Y. Li, J. Ren, H. Fu, and G. Sun, "GASSM: Global attention and state space model based end-to-end hyperspectral change detection," *J. Franklin Inst.*, vol. 362, Feb. 2025, Art. no. 107424.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:42:48 UTC from IEEE Xplore. Restrictions apply.

[29] S. Liu, L. Bruzzone, F. Bovolo, M. Zanetti, and P. Du, "Sequential spectral change vector analysis for iteratively discovering and detecting multiple changes in hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 8, pp. 4363–4378, Aug. 2015.
[30] S. Achour, M. Chikr Elmezouar, N. Taleb, K. Kpalma, and J. Ronsin, "A PCA-PD fusion method for change detection in remote sensing multi temporal images," *Geocarto Int.*, vol. 37, no. 1, pp. 196–213, 2022.
[31] A. A. Nielsen, K. Conradsen, and J. J. Simpson, "Multivariate alteration detection (MAD) and MAF postprocessing in multispectral, bitemporal image data: New approaches to change detection studies," *Remote Sens. Environ.*, vol. 64, no. 1, pp. 1–19, 1998.
[32] F. Bovolo and L. Bruzzone, "A theoretical framework for unsupervised change detection based on change vector analysis in the polar domain," *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 1, pp. 218–236, Jan. 2007.
[33] A. A. Nielsen, "The regularized iteratively reweighted MAD method for change detection in multi- and hyperspectral data," *IEEE Trans. Image Process.*, vol. 16, no. 2, pp. 463–478, Feb. 2007.
[34] T. Habib, J. Inglada, G. Mercier, and J. Chanussot, "Support vector reduction in SVM algorithm for abrupt change detection in remote sensing," *IEEE Geosci. Remote Sens. Lett.*, vol. 6, no. 3, pp. 606–610, Jul. 2009.
[35] H. Mahdipour, A. Sharifi, M. Sookhak, and C. R. Medrano, "Ultrafusion: Optimal fuzzy fusion in land-cover segmentation using multiple panchromatic satellite images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 5721–5733, 2024.
[36] X. Li et al., "A cross-domain coupling network for semantic segmentation of remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.
[37] X. Li, F. Xu, A. Yu, X. Lyu, H. Gao, and J. Zhou, "A frequency decoupling network for semantic segmentation of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5607921.
[38] Q. Wang, Z. Yuan, Q. Du, and X. Li, "GETNET: A general end-to-end 2-D CNN framework for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 3–13, Jan. 2019.
[39] Y. Huang et al., "Cross-scene wetland mapping on hyperspectral remote sensing images using adversarial domain adaptation network," *ISPRS J. Photogramm. Remote Sens.*, vol. 203, pp. 37–54, Sep. 2023.
[40] J. Qu, S. Hou, W. Dong, Y. Li, and W. Xie, "A multilevel encoder–decoder attention network for change detection in hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5518113.
[41] J. Ding, X. Li, and L. Zhao, "CDFormer: A hyperspectral image change detection method based on transformer encoders," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.
[42] Y. Wang et al., "Spectral–spatial–temporal transformers for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5536814.
[43] Z. Wang, F. Gao, J. Dong, and Q. Du, "Global and local attention-based transformer for hyperspectral image change detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 22, pp. 1–5, 2025.
[44] M. Hu, C. Wu, and L. Zhang, "GlobalMind: Global multi-head interactive self-attention network for hyperspectral change detection," *ISPRS J. Photogramm. Remote Sens.*, vol. 211, pp. 465–483, May 2024.
[45] S. Woo, J. Park, J.-Y. Lee, and I. S. Kweon, "CBAM: Convolutional block attention module," in *Proc. Eur. Conf. Comput. Vis.*, Sep. 2018, pp. 3–19.
[46] Z. Yu et al., "Searching central difference convolutional networks for face anti-spoofing," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Aug. 2020, pp. 5295–5305.
[47] Z. Chen, Z. He, and Z.-M. Lu, "DEA-Net: Single image dehazing based on detail-enhanced convolution and content-guided attention," *IEEE Trans. Image Process.*, vol. 33, pp. 1002–1015, 2024.
[48] D. Chen, X. Liang, L. Wang, Q. Guo, and J. Zhang, "Global difference-aware mamba for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5523214.
[49] S. Xu et al., "Strengthened residual graph and multiscale gated guided convolutional fusion network for hyperspectral change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5540014.
[50] R. Song, W. Ni, W. Cheng, and X. Wang, "CSANet: Cross-temporal interaction symmetric attention network for hyperspectral image change detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.
[51] X. Wang, K. Zhao, X. Zhao, and S. Li, "GTMSiam: Gated transmitting-based multiscale Siamese network for hyperspectral image change detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.
[52] L. Wu, J. Peng, B. Yang, W. Sun, and Z. Ye, "AIWSEN: Adaptive information weighting and synchronized enhancement network for hyperspectral change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5503412.



Zhenyang Xie (Graduate Student Member, IEEE) received the B.E. degree in software engineering from Jiangxi Normal University, Nanchang, China, in 2024. He is currently pursuing the M.E. degree with the School of Information Engineering, Jiangxi University of Water Resources and Electric Power, Nanchang.

His research interests include deep learning and image processing in remote sensing.



Li Lv received the Ph.D. degree from Hohai University, Nanjing, China, in 2019.

Currently, she is a Professor with the School of Information Engineering, Jiangxi University of Water Resources and Electric Power, Nanchang, China. Her research interests include evolutionary computation, image processing, pattern recognition, and big data analysis.



Hongmin Gao (Senior Member, IEEE) received the B.S. degree in communication engineering and the Ph.D. degree in computer application technology from Hohai University, Nanjing, China, in 2006 and 2014, respectively.

He is currently a Professor with the College of Computer Science and Software Engineering, Hohai University. In the last five years, he has undertaken two research projects supported by the National Natural Science Foundation of China (NSFC), one research project supported by Jiangsu Natural Science Foundation, one research project supported by the transformation of scientific and technological achievements in Jiangsu Province, and participated in the National Key Research and Development Program of China. His research interests include deep learning, information fusion, and image processing in remote sensing.

Dr. Gao part of the research results have been applied in water resource flood disaster monitoring and operation decision supporting systems of water conservancy, which won the second prize in Jiangxi Province Science and Technology Advancement Reward in 2018.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:42:48 UTC from IEEE Xplore. Restrictions apply.



Shufang Xu (Member, IEEE) received the B.S. degree in applied electronic technology from Anhui Normal University, Wuhu, China, in 2002, and the master's and Ph.D. degrees in communication engineering from Nanjing University of Aeronautics and Astronautics, Nanjing, China, in 2005 and 2017, respectively.

In the last 5 years, she has participated in two research projects supported by the National Key Research and Development Program of China and one research project supported by the Jiangsu Province Key Research and Development Program. She is currently an Associate Professor with the College of Computer Science and Software Engineering, Hohai University, Nanjing. Her research interests include uncrewed aerial vehicle (UAV) swarm cooperation, information fusion, and image processing in remote sensing.

Dr. Xu part of the research results have been applied in water resource flood disaster monitoring and the operation decision-supporting system of water conservancy, which won the China Industry-University-Research Cooperation Innovation Award in 2019 and 2021.



Jeng-Shyang Pan (Senior Member, IEEE) received the Bachelor of Science degree from the Department of Electronic Engineering, National Taiwan University of Science and Technology, Taipei, Taiwan, in 1986, the master's degree from the Department of Communications Engineering, National Chiao Tung University, Hsinchu, Taiwan, in 1988, and the Ph.D. degree in electrical engineering from the University of Edinburgh, Edinburgh, U.K., in 1996.

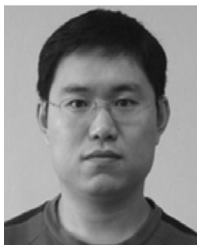
He is currently a Professor with the School of Artificial Intelligence, Nanjing University of Information Science and Technology, Nanjing, China, and an Adjunct Professor with the College of Computer Science and Engineering, Shandong University of Science and Technology, Qingdao, China. His research interests include soft computing, information security, and signal processing.

Dr. Pan is an IET Fellow, U.K., and has been the Vice Chair of the IEEE Tainan Section.



Wenping Chen received the Ph.D. degree from Hohai University, Nanjing, China, in 2025.

He is currently a Lecturer with Jiangxi University of Water Resources and Electric Power, Nanchang, China. His research interests include deep learning and intelligent optimization algorithms.



Longzhe Han received the Ph.D. degree from Korea University, Seoul, South Korea, in 2013.

He is currently a Lecturer with the School of Information Engineering, Jiangxi University of Water Resources and Electric Power, Nanchang, China. His research interests include cognitive radio networks, future Internet, network security, multimedia communications, machine-to-machine communications, and fifth-generation heterogeneous networks.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:42:48 UTC from IEEE Xplore. Restrictions apply.